

Entropy in Optimal Transport

Exact OT

Given the Optimal transport problem

$$\underline{L_c(a,b) = \min_{P \in U(a,b)} \langle C, P \rangle = \min_{P \in U(a,b)} \sum_{i,j} C_{ij} P_{ij}}$$

$L_c \rightarrow$ result of minimization

$P \rightarrow$ Transport Plan

$C \rightarrow$ Cost table

Find the cheapest way to ship supply a (source $\sim u$) to demand b (target $\sim v$),
given a cost table C and plan P

$U(a,b) \rightarrow$ feasible set (valid plan)

which is a set of non-negative matrices
(every legal way to move mass)

The cheapest plan is sharp & sparse.

Fundamental fact of Linear Programming

Minimizing a linear function over a convex set defined by linear constraints, the optimum is guaranteed to exist at a corner or vertex of the shape.

The Problem

Exact OT is accurate but slow and a combinatorial search problem.

Idea: Blur the plan P so the soln is smooth instead of a hard corner.

Entropy

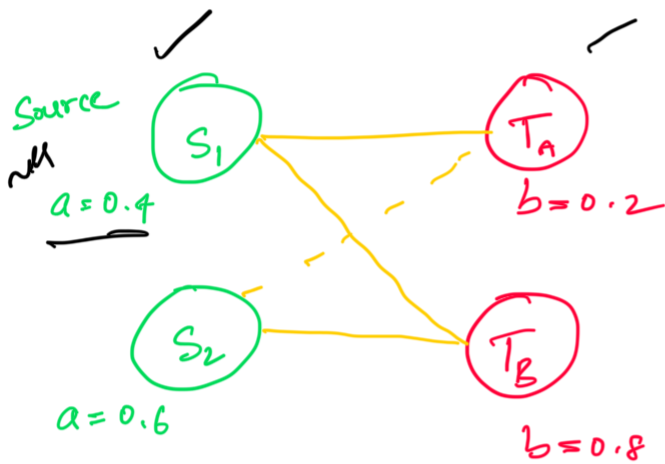
$$H(P) = - \sum_{i,j} P_{ij} (\log P_{ij} - 1)$$

$H \rightarrow$ Scalar quantity (measures blurriness)

$H \sim 0$ (sharp)

Example

Given sources S_1, S_2 ($m_1 = 10, m_2 = 15$) and target T_A, T_B ($m_A = 5, m_B = 20$)



$$P = \begin{matrix} & T_A & T_B \\ S_1 & \begin{bmatrix} 0.2 & 0.2 \end{bmatrix} \\ S_2 & \begin{bmatrix} 0 & 0.6 \end{bmatrix} \end{matrix} \cdot a$$

$$H(P) = 1.95 \quad \text{(spread score)}$$

$$b \begin{bmatrix} 0.1 & 0.3 \\ 0.1 & 0.5 \end{bmatrix}$$

$$H = 2.17$$

$(a, b) \rightarrow$ Probability measures

$P \rightarrow$ How much fraction of mass

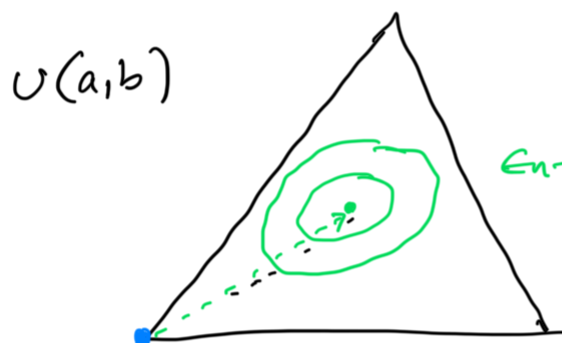
is moved across different routes

Adding $-\epsilon H$ to the original eqn

At $\epsilon = 0$, the optimum is pinned to a corner of the triangle. ' $-\epsilon H$ ' changes the triangle into smooth bowl

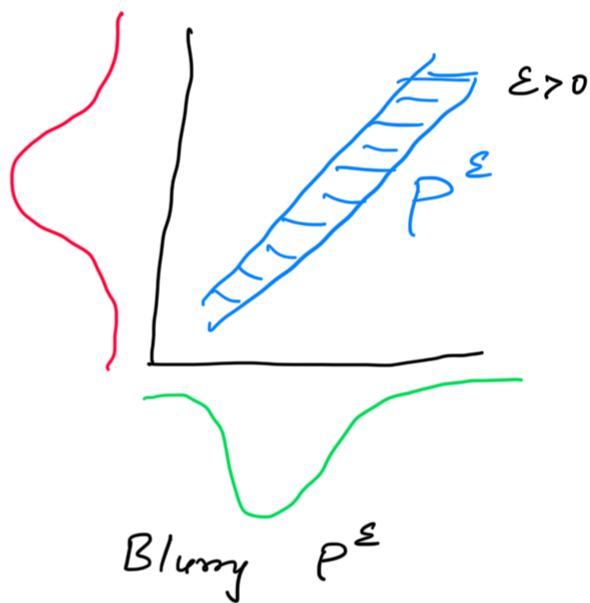
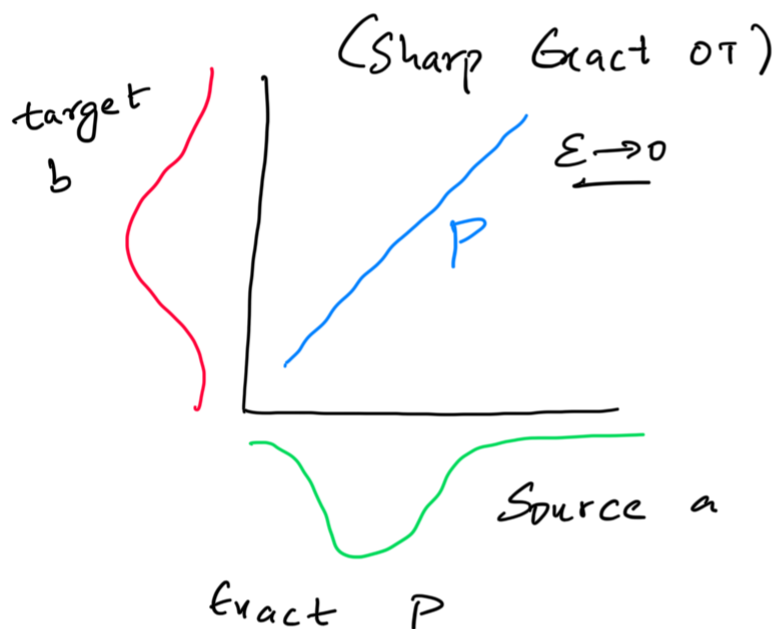
P, P^ϵ

$$\mathcal{L}_C^\epsilon(a, b) = \min_{P \in U(a, b)} \langle C, P \rangle - \epsilon H(P)$$



Exact OT ($\epsilon \rightarrow 0$)

True Optimum



Why the Optimal P^ϵ is guaranteed & Unique.

$$H(P) = - \sum P_{ij} (\log P_{ij} - 1) -$$

$$\partial H / \partial P_{ij} = -\log P_{ij}$$

$$\partial^2 H / \partial P_{ij}^2 = -\frac{1}{P_{ij}} \quad (\text{curvature})$$

$$\text{Hessian} = -\text{diag} \left(1 / P_{ij} \right)$$

The curvature is negative everywhere
 $(\because P_{ij} > 0)$.

$\rightarrow H$ is concave

which means $-EH$ is strictly convex
(bowl shape)

So, There is exactly one best plan P^E
 (at lowest point) and that is guaranteed
 and Unique.

The Trade-off

$$\checkmark \quad \langle C, P^\varepsilon \rangle \geq \langle C, P \rangle = L_c(a, b)$$

blurred Plan
(parallel sinkhorn)

Actual Plan
(Combinatorial Search)

Sinkhorn's Algorithm

Given,

$$L_c^\varepsilon(a, b) = \min_{P \in U(a, b)} \langle C, P \rangle - \varepsilon H(P)$$

which is strictly convex, so it has
one unique P^ε sitting in the smooth
interior of $U(a, b)$

Shape awareness $\rightarrow$ Total Cost + reg (bowl)

Setup the problem with Lagrange
multipliers (dual variables)

f_i $\rightarrow$ vector for row constraint
 $(P_{i1} = a)$

$\underline{g_j} \rightarrow$ vector for col constraint
 $(P1 = b)$

$$\underline{\varepsilon(P, f, g)} = \underline{\langle C, P \rangle - \varepsilon_H(P) - \langle f, P1_m - a \rangle - \langle g, P1_n^T - b \rangle}$$

$f \in \mathbb{R}^1, g \in \mathbb{R}^n$ (marginal constraints)

$$C_{ij} + \varepsilon \log P_{ij} - f_i - g_j = 0$$

($\therefore$ Slope = 0)

$$\underline{P_{ij} = e^{(f_i + g_j - C_{ij})/\varepsilon}} \rightarrow P_{ij} = \underbrace{e^{(f_i/\varepsilon)}}_{\text{row}} \cdot \underbrace{e^{(g_j/\varepsilon)}}_{\text{col}} \cdot \underbrace{e^{-C_{ij}/\varepsilon}}_{\text{cost}}$$

$$P_{ij} = \text{diag}(u) \cdot K \cdot \text{diag}(v)$$

$K = e^{-C/\varepsilon}$ (desirability pushes cheap notes to 1 & exp 0)

$$P1_n = a$$

$$, \quad P1_m = b$$

$$u \odot (Kv) = a$$

$$v \odot (K^T u) = b$$

$$\underline{u = \frac{a}{Kv}}$$

$$\underline{v = \frac{b}{K^T u}}$$

(Two eqns, Two unknowns)

We solve them by alternating (u needs v ,

v needs u)

Sinkhorn

$$K = \exp(-C/\epsilon)$$

$$v = \text{ones}$$

repeat:

$$\underline{u} \leftarrow a / (K v) \quad \# \text{ row-sum} = \underline{a}$$

$$\underline{v} \leftarrow b / (K^T u) \quad \# \text{ col-sum} = \underline{b}$$

$$\text{return } \underline{P} = \text{diag}(u) K \text{diag}(v)$$

Why it works?

There exists a u, v pair that fixes both margins because $K = e^{-C/\epsilon}$ is strictly positive (entries).

This hypothesis is also known as

Sinkhorn's theorem.

P gets updated alternatively (fix row $\rightarrow$ fix col $\rightarrow$ fix row $\rightarrow$ fix col $\rightarrow \dots$) eventually

stops as they converge. (like oscillation)